SUMMER EXAMINATION 2026

Machine Learning - Version B

BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: ____________________ Roll No.: ____________________ Date: ____________________

1. The ROC AUC summarizes ranking quality across
(A) classification thresholds (B) learning rates only
(C) cluster counts (D) feature encodings

2. L1 regularization encourages
(A) sparse coefficients (B) large weights
(C) more clusters (D) higher variance

3. In Q-learning, Q(s,a) estimates expected return for
(A) an action in a state (B) a feature value
(C) a class label (D) a cluster

4. An epsilon-greedy policy explores randomly with probability
(A) epsilon (B) one minus epsilon always
(C) the learning rate only (D) the discount factor only

5. Recall equals TP divided by
(A) TP plus FN (B) TP plus FP
(C) TN plus FN (D) all predictions

6. Underfitting often indicates a model is
(A) too simple (B) too complex
(C) leaking data (D) using too many labels

7. A 2 by 2 confusion matrix contains
(A) four outcome cells (B) two models
(C) two features (D) four classes necessarily

8. Overfitting is associated with low training error and high
(A) validation error (B) training accuracy only
(C) bias only (D) regularization strength

9. Accuracy can be misleading when classes are
(A) imbalanced (B) balanced
(C) continuous (D) standardized

10. A CNN convolution filter detects
(A) local spatial patterns (B) only class priors
(C) database keys (D) sequence rewards

11. For TP=12 and FN=3, recall is
(A) 0.80 (B) 0.20
(C) 0.75 (D) 0.60

12. Backpropagation computes gradients using the
(A) chain rule (B) Bayes rule only
(C) Markov rule (D) Pythagorean theorem

13. Mean squared error averages
(A) squared residuals (B) absolute labels
(C) class probabilities only (D) feature counts

14. Bagging trains models on
(A) bootstrap samples (B) one fixed sample only
(C) ordered labels (D) future data

15. Unsupervised learning searches for
(A) structure without labels (B) known class names
(C) payment status (D) manual rules

16. A discount factor near zero emphasizes
(A) immediate rewards (B) distant rewards
(C) feature scaling (D) test accuracy

17. Supervised learning requires
(A) labeled examples (B) only unlabeled data
(C) no target variable (D) random rewards

18. Boosting builds an ensemble
(A) sequentially focusing on errors (B) in parallel with no interaction
(C) from one tree (D) only from test data

19. The learning rate controls
(A) step size in optimization (B) number of classes
(C) test proportion only (D) feature count

20. Missing numeric values may be imputed with the
(A) median (B) class name only
(C) file path (D) gradient sign

21. A perceptron is a
(A) single linear threshold unit (B) clustering algorithm
(C) tree ensemble (D) language model

22. A larger K in KNN usually makes the decision boundary
(A) smoother (B) more jagged
(C) undefined (D) linear always

23. A reproducible pipeline fixes seeds and records
(A) data, code, and configuration (B) only accuracy
(C) only screenshots (D) only labels

24. A hyperparameter is set
(A) before or outside parameter fitting (B) only by backpropagation
(C) by the test label (D) after deployment only

25. Min-max scaling typically maps values to
(A) a chosen interval such as 0 to 1 (B) integer labels
(C) random signs (D) probabilities always

26. Reinforcement learning uses actions, states, and
(A) rewards (B) SQL joins
(C) parse trees (D) static labels only

27. DBSCAN identifies dense regions using neighborhood radius and
(A) minimum points (B) tree depth
(C) class weights (D) learning rate

28. An epoch is one pass through the
(A) training data (B) test labels only
(C) hyperparameter grid (D) confusion matrix

29. K-means alternates assignment and
(A) centroid update (B) tree pruning
(C) gradient clipping (D) label encoding

30. An activation function adds
(A) nonlinearity (B) database indexing
(C) data leakage (D) a test split

31. Machine learning learns patterns from data to make
(A) predictions or decisions (B) only web pages
(C) fixed hand-coded outputs (D) network cables

32. Entropy measures
(A) impurity or uncertainty (B) distance to origin
(C) learning rate (D) number of layers

33. Hierarchical clustering produces a
(A) dendrogram (B) confusion matrix
(C) ROC curve (D) learning curve only

34. MAE averages the absolute
(A) prediction errors (B) squared features
(C) class labels (D) gradient steps

35. The residual in regression is actual value minus
(A) predicted value (B) feature mean
(C) class prior (D) tree depth

36. Data leakage occurs when information from the future or target
(A) enters training improperly (B) is normalized
(C) is shuffled (D) is missing

37. An RNN processes inputs while maintaining a
(A) hidden state (B) decision stump
(C) centroid (D) confusion table

38. A lower validation loss generally indicates
(A) better validation fit (B) more leakage necessarily
(C) higher bias always (D) fewer samples only

39. Gradient descent moves parameters opposite the
(A) gradient of loss (B) data mean
(C) test set (D) class label

40. KNN predicts using the labels of nearby
(A) training points (B) decision nodes
(C) principal components only (D) random samples

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1. PCA finds orthogonal directions ordered by
(A) explained variance (B) class frequency
(C) reward (D) accuracy only

2. A test set should be held out for
(A) final generalization evaluation (B) feature creation
(C) repeated tuning (D) data collection only

3. RMSE is the square root of
(A) MSE (B) MAE
(C) R squared (D) variance only

4. Logistic regression commonly outputs
(A) a class probability (B) a cluster centroid
(C) a parse tree (D) a continuous image only

5. Support vectors are training points closest to the
(A) decision boundary (B) centroid
(C) root node (D) mean label

6. R squared measures variance explained relative to a
(A) baseline model (B) confusion matrix
(C) cluster center (D) learning rate

7. The target in a regression problem is usually
(A) continuous (B) a cluster ID only
(C) a token sequence (D) a Boolean rule only

8. Feature selection removes
(A) unhelpful input variables (B) all target labels
(C) test cases (D) model predictions

9. Model drift means the data or relationship changes after
(A) deployment (B) normalization
(C) training split (D) label encoding

10. A decision tree splits data using
(A) feature tests (B) random rewards
(C) vector norms only (D) SQL transactions

11. L2 regularization penalizes the
(A) sum of squared weights (B) number of classes
(C) absolute residual only (D) test size

12. The first principal component has the greatest
(A) variance (B) mean label
(C) test error (D) number of samples

13. Linear regression models a target as a
(A) weighted linear combination (B) tree depth
(C) cluster assignment (D) probability table only

14. The sigmoid function maps real values to
(A) between 0 and 1 (B) between -1 and 1 only
(C) all integers (D) unit vectors only

15. Stratified splitting preserves
(A) class proportions (B) feature means exactly
(C) time order always (D) duplicate rows

16. A kernel can enable SVMs to model
(A) nonlinear boundaries (B) missing values
(C) test leakage (D) class names

17. One-hot encoding represents a category with
(A) binary indicator columns (B) a single ordered rank
(C) a text paragraph (D) a distance matrix only

18. Naive Bayes assumes features are conditionally independent given the
(A) class (B) distance
(C) tree depth (D) reward

19. Gradient boosting fits new learners to
(A) current residual errors (B) feature names
(C) random labels (D) test predictions only

20. Cross-validation repeatedly splits data into training and
(A) validation folds (B) test labels only
(C) features and targets permanently (D) random classes

21. Fairness evaluation checks whether outcomes differ across
(A) relevant groups (B) random file names
(C) feature columns only (D) epochs only

22. A feature is an input variable used by a
(A) model (B) compiler
(C) database index only (D) test runner

23. Standardization commonly gives a feature mean 0 and
(A) standard deviation 1 (B) range 1
(C) median 1 (D) variance 0

24. Early stopping can reduce overfitting by halting when validation performance
(A) stops improving (B) reaches zero training loss
(C) has more features (D) uses no labels

25. Random forests add randomness by selecting subsets of
(A) features at splits (B) test labels
(C) reward values (D) output classes only

26. An outlier is an observation that is
(A) unusually distant from typical values (B) always incorrect
(C) a missing label (D) a duplicate key only

27. Grid search evaluates
(A) specified hyperparameter combinations (B) all possible datasets
(C) only one model (D) random gradients

28. Precision equals TP divided by
(A) TP plus FP (B) TP plus FN
(C) TN plus FP (D) all negatives

29. Explainability helps stakeholders understand
(A) model predictions (B) network cables
(C) database backups (D) file compression

30. Pooling usually reduces
(A) spatial resolution (B) number of labels
(C) input types (D) training epochs only

31. The K in K-means is the number of
(A) clusters (B) features
(C) iterations only (D) classes in the test set

32. An SVM seeks a separating hyperplane with a large
(A) margin (B) entropy
(C) learning rate (D) cluster count

33. Pruning a tree helps reduce
(A) overfitting (B) missing values
(C) class count (D) feature scaling

34. LSTM gates regulate information in the
(A) cell state (B) test set
(C) feature scaler (D) tree root

35. A training set is used to
(A) fit model parameters (B) choose final policy only
(C) report unbiased test error (D) encrypt records

36. ReLU outputs max of zero and
(A) its input (B) its square
(C) its class (D) its probability

37. Specificity is the true negative rate, TN divided by
(A) TN plus FP (B) TN plus FN
(C) TP plus FP (D) all positives

38. Monitoring should track performance, latency, and
(A) input or output drift (B) font size
(C) source comments (D) tree names

39. Bayes theorem relates posterior, likelihood, prior, and
(A) evidence (B) variance only
(C) gradient (D) residual

40. If a classifier gets 90 correct out of 100, accuracy is
(A) 0.90 (B) 0.10
(C) 0.50 (D) 90.0